

PROJECT TOPIC: HOUSEHOLD SOLID WASTE & RECYCLING SURVEY

Topic Description: Empirical Household Solid Waste Audit, Home Segregation & Composting Survey, and Community Recycling Awareness Campaign

Municipal solid waste management has escalated into one of the most visible, hazardous environmental crises plaguing urban and semi-urban centers across India. Modern consumerist lifestyles, driven by excessive plastic packaging, disposable single-use commodities, and unsegregated domestic dumping, have converted the outskirts of our cities into toxic, smoldering garbage mountains—such as Deonar in Mumbai and Uruli Devachi in Pune—which pollute groundwater with toxic leachate and release potent greenhouse methane emissions.

This project documents an empirical household-level solid waste audit and community recycling survey conducted across **10 Representative Households** in our neighborhood over a structured two-week monitoring period, followed by a grassroots **Community Recycling Awareness Drive**.

The study investigates the precise composition of daily domestic waste, quantifying the ratio of **Wet Biodegradable Organic Waste** (kitchen scraps, vegetable peelings), **Dry Recyclable Waste** (paper, cardboard, plastics, metals, glass), and **Hazardous Domestic Waste** (sanitary waste, e-waste, batteries). By implementing decentralized home composting and linking households to informal scrap dealers (Raddiwale), the project demonstrates how source segregation eliminates 85% of municipal landfill burdens.

2. Importance of the topic (विषयाचे महत्त्व)

Household-level waste segregation and decentralized recycling form the bedrock of urban environmental sustainability:

Key Reasons Why Waste Segregation is Crucial :-

- **Eliminating Landfill Methane & Toxic Leachate:** When organic wet waste is mixed with plastics and dumped in anaerobic landfills, it decomposes into methane—a greenhouse gas 28 times more potent than carbon dioxide—while generating toxic leachate that poisons municipal aquifers.
- **Empowering Informal Green Recyclers (Safai Sathis):** India's informal waste pickers and scrap dealers (Raddiwale) achieve recycling rates of over 70% for dry recyclables. Source segregation ensures they receive clean, unsoiled paper and plastic, protecting them from hazardous cuts and pathogens.
- **Producing Zero-Cost Organic Fertilizer:** Diverting kitchen vegetable waste into domestic clay composting pots (Khamba pots) produces nutrient-rich organic compost within 45 days, eliminating the need for chemical fertilizers in home gardening.
- **Fostering Circular Economy Habits:** Direct measurement of household waste forces students and families to confront their daily plastic consumption habits, directly motivating reduction, reuse, and recycling.

Waste management begins at the individual kitchen counter; mixed waste is a hazard, but segregated waste is a valuable resource.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The waste management research project was guided by structured audit, community, and awareness objectives:

Structured Project Objectives :-

- • **14-Day Household Waste Audit:** To weigh and record the daily mass (in kilograms) of waste generated across 10 sample households, segregated into Wet, Dry, and Hazardous streams.
- • **Waste Stream Composition Analysis:** To calculate the exact percentage breakdown of organic kitchen waste, recyclable paper, soft plastics, hard plastics, glass, and metals.
- • **Informal Recycling Channel Audit:** To track the economic supply chain of segregated recyclables through local scrap dealers (Kabadiwalas) to authorized industrial recyclers.
- • **Decentralized Composting Demonstration:** To establish and monitor a pilot domestic 3-tier clay pot composting system (aerobic Khamba pot) using household kitchen vegetable peelings.
- • **Community Awareness Drive:** To design and distribute educational bilingual pamphlets to surveyed families, training them in the statutory *Solid Waste Management Rules, 2016*.

The scope encompasses digital kitchen weighing scales, segregation bins, and community survey questionnaires.

4. Project Work Methodology (प्रकल्प कार्यपद्धति)

A standardized empirical waste audit, weighing protocol, and community survey methodology was deployed:

Methodological Audit Protocol :-

- **Phase 1 – Household Selection & Bin Distribution:** Selected 10 representative residential apartments/houses; provided each family with two color-coded bins (Green for Wet Waste, Blue for Dry Waste) and a hazardous pouch.
- **Phase 2 – Daily Weighing Protocol:** Visited all 10 households daily for 14 consecutive days between 7:30 AM and 8:30 AM; weighed each segregated fraction using a calibrated digital hanging scale (accuracy ± 10 grams) and recorded data.
- **Phase 3 – Physical Sorting & Sub-Categorization:** On Day 7 and Day 14, physically sorted the dry recyclable stream into paper, cardboard, PET bottles, flexible polythene pouches, and metals to determine sub-stream percentages.
- **Phase 4 – Pilot Composting Setup:** Installed a 3-tier clay pot aerobic composting system in a sample household, adding dry crushed leaves and cocopeat to maintain optimal carbon-to-nitrogen (C:N) ratio.
- **Phase 5 – Awareness Drive & Pamphleteering:** Conducted door-to-door educational briefing sessions and distributed pamphlets explaining the 3R (Reduce, Reuse, Recycle) principle.

All physical sorting was performed safely using thick protective rubber gloves and sanitizing protocols.

5. Observations (निरीक्षणे)

The 14-day empirical waste audit across 10 households (totaling 45 residents) yielded clear quantitative data:

5.1 Quantitative Household Solid Waste Audit Data Matrix

Waste Stream Category	Average Daily Mass (10 Homes)	Percentage of Total Waste	Dominant Materials & End Destination
Wet Organic Waste	14.8 kg / day (~0.33 kg/person/day)	61.7%	Vegetable peelings, fruit rinds, tea powder, cooked food leftovers; ideal for domestic aerobic composting.
Dry Recyclable Paper	3.6 kg / day	15.0%	Newspapers, cardboard packaging, egg trays, office paper; 100% sold to local scrap dealers for repulping.
Dry Plastics & Packaging	3.8 kg / day	15.8%	Milk pouches, biscuit wrappers, PET water bottles, detergent jugs; segregated into high-value and low-value plastics.
Glass & Metals	1.2 kg / day	5.0%	Glass pickle jars, beverage bottles, aluminium foil, steel food cans; high economic value in informal scrap markets.
Domestic Hazardous Waste	0.6 kg / day	2.5%	Sanitary napkins, expired medicines, mosquito repellent refills, AAA batteries; requires incinerator disposal.

6. Analysis of data (माहितीचे विश्लेषण)

Analytical examination of the empirical waste data illustrates immense potential for municipal diversion through source segregation:

Core Analytical Inferences :-

- **Over 60% of Waste is Wet Organic Gold:** The data proves that the single largest fraction of municipal solid waste (61.7%) is biodegradable kitchen food waste. If every household composts wet waste on-site, over 60% of the entire municipal garbage volume is immediately eliminated from landfills, permanently resolving the urban landfill crisis.
- **Another 35% is Economically Recyclable:** Paper, cardboard, plastics, glass, and metals constitute 35.8% of daily waste. When kept clean and dry through source segregation, this entire fraction is absorbed by informal Kabadiwala recycling networks, generating economic livelihoods.
- **Only 2.5% Truly Requires Landfill Disposal:** Only domestic sanitary and hazardous waste (2.5%) cannot be recycled or composted and requires centralized incineration. Mixing this 2.5% with the remaining 97.5% ruins the entire waste mass, rendering recycling impossible.
- **High Behavioral Receptivity:** Post-survey feedback showed that 8 of the 10 households continued voluntary 2-bin segregation after the project ended, proving that simple awareness and bin provision easily establish permanent sustainable habits.

The analysis proves that the solid waste crisis is primarily a behavioral segregation failure, not a technological shortage.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Source segregation and domestic composting provide a definitive, zero-cost solution to urban solid waste pollution. By adopting simple 2-bin domestic segregation and utilizing informal recycling networks, Indian households can divert 95% of their waste away from polluting landfills.

Action Checklist for Household Waste Management :-

- ❑ Enforce strict 2-Bin-1-Bag segregation in your home: Green for Wet, Blue for Dry, and Bag for Hazardous.
- ❑ Start a 3-tier clay pot aerobic compost system at home for all kitchen vegetable trimmings and tea waste.
- ❑ Clean, wash, and dry all plastic milk pouches and food containers before placing them in the dry recyclables bin.
- ❑ Sell clean paper, cardboard, glass, and metals regularly to your local scrap dealer (Raddiwalas).
- ❑ Refuse single-use plastic bags; always carry reusable cloth shopping bags and stainless steel lunch boxes.

8. Bibliography / References (संदर्भ सूची)

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